rag -- 2026-11-09

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P4] SkinMind: Equipping Agentic RAG with Human-Inspired Graph Probing for Reliable Dermatology AI -- SkinMind integrates interactive graph probing, multimodal retrieval, and token-budget management, delivering the largest reported performance gains and illustrating a comprehensive agentic RAG architecture worth studying in depth.

Themes

Several papers explore how richer retrieval mechanisms affect downstream generation quality. P1 shows that a two-tier compacted memory can match or exceed transcript-based RAG while using 34 % fewer prompt tokens, but its advantage disappears on adversarial or longer-answer queries, highlighting a trade-off between compactness and grounding fidelity [P1]. P2 demonstrates that similarity-based example retrieval modestly lifts code-generation success (pass@1 +2.5 pts) over zero-shot and random baselines, confirming that relevance of retrieved items matters even in algorithmic problem-solving [P2]. P3 and P4 both move beyond flat text retrieval to graph-structured evidence: GRAC retrieves graph-caption pairs to improve surgical scene description (CIDEr-D +1.65) but gains plateau after three examples, while SkinMind employs an agent that iteratively probes a multimodal knowledge graph, achieving ~8-point VQA improvements over static RAG baselines [P3][P4]. P5 takes a different route by eliminating learned retrievers altogether, training a multilingual search agent that issues executable commands; this reduces redundant searches by 70 % and improves exact-match accuracy (+3.6 pts) compared with other DCI systems, suggesting that search-policy learning can replace similarity ranking in cross-lingual settings [P5]. Across the set, a recurring evaluation gap is the reliance on narrow, domain-specific benchmarks (LoCoMo, CodeContests, EndoVis-18, dermatology VQA, three multilingual tasks), making it hard to compare the generality of these retrieval designs.

Themes could not be fact-checked (HTTPError).

At a glance

[P1] Compaction, decay and reinforcement in a two-tier conversational memory: a paired evaluation against retrieval-augmented generation · relevance 0.43
The paper compares a compacting two-tier conversational memory to transcript-based retrieval-augmented generation on the LoCoMo benchmark, finding modest overall gains but strong advantages on temporal questions and a loss on adversarial ones.

[P2] Assessing the Impact of Retrieval-Augmented Generation on Algorithmic Problem-Solving · relevance 0.38
Retrieval-augmented generation (RAG) using similarity-based example retrieval modestly improves LLM performance on competitive programming, raising pass@1 to 20.37% and Elo to 1014 versus weaker baselines.

[P3] GRAC: Graph-Based Retrieval-Augmented Captioning for Surgical Scene Description · relevance 0.38
GRAC is a graph-based, retrieval-augmented captioning system that outperforms prior surgical captioning models on standard benchmarks, but its gains rely on specific components and its evaluation is limited to the datasets and protocols used.

[P4] SkinMind: Equipping Agentic RAG with Human-Inspired Graph Probing for Reliable Dermatology AI · relevance 0.37
SkinMind is an agentic retrieval-augmented generation system that interactively probes a large multimodal

dermatology knowledge graph to improve VQA performance, yielding ~8-point gains over direct answering.

[P5] MARS: Training Multilingual Search Agents for Retriever-Free Evidence Acquisition · relevance 0.33
MARS is a reinforcement-learning-based multilingual search agent that interacts directly with raw language-specific corpora via executable commands, achieving higher exact-match accuracy while markedly cutting redundant searches compared with retrieval-augmented baselines.

[P1] Compaction, decay and reinforcement in a two-tier conversational memory: a paired evaluation against retrieval-augmented generation

Osamudiamen Edogun, Joshua Obigwe, Irfan Mehmood · arXiv 2611.09807v1 · https://arxiv.org/abs/2611.09807v1 · relevance 0.43

TL;DR: The paper compares a compacting two-tier conversational memory to transcript-based retrieval-augmented generation on the LoCoMo benchmark, finding modest overall gains but strong advantages on temporal questions and a loss on adversarial ones.

Problem
Conversational agents must decide which parts of a dialogue to retain, balancing memory size against answer quality; the authors frame this as a trade-off between compacting turns into atomic facts and retrieving the full transcript [1]. Prior work suggests placing less information in the prompt improves accuracy, motivating the compacting approach [2].

Method
The authors implement a two-tier memory that extracts atomic facts, applies decay and reinforcement, and optionally hydrates raw turns; the default renders up to 20 facts and 5 hydrated turns (~32 tokens per fact, ~66 tokens per hydrated turn) [3].

Key results
- On the four categories reported by comparable work, compaction achieves 71.4 % accuracy versus 64.6 % for the baseline, a 6.8-point lead (p = 5.5e-08) while using 34 % fewer prompt tokens [1].
- Including the adversarial category reduces the overall lead to 2.0 points, which is not statistically significant [1].

- Per-conversation accuracy ranges from 56.4 % to 73.2 % for compaction and 55.2 % to 71.1 % for the baseline [4].
Passing the baseline consumes 1,298 tokens, while the fact list alone uses 967 tokens [5].
- Latency: paired runs show mean latencies of 2,721 ms (compaction) vs 2,295 ms (baseline); median latency is 28 ms faster for compaction, with indistinguishable performance up to the 90th percentile [5].
- Four negative results are reported (decay weight, archival search gate, list-shape gate, answer-shape clauses) with no measurable benefit [5].

Limitations
The advantage disappears when the adversarial category is included, indicating a loss on wordy, phrasing-heavy answers [1][4]. Performance gains decline as the reference answer requires more source wording (lead of 9.6 points with 1-2 words, dropping to -3.1 points beyond eight words) [1]. Single-hop improvements are not statistically powered, and the evaluation uses the synthetic LoCoMo benchmark scored by a model judge [2][4].

Sources

[1] arXiv 2611.09807v1.pdf, arXiv abstract

[2] arXiv 2611.09807v1.pdf, page 2

[3] arXiv 2611.09807v1.pdf, page 4

[4] arXiv 2611.09807v1.pdf, page 5

[5] arXiv 2611.09807v1.pdf, page 7

[P2] Assessing the Impact of Retrieval-Augmented Generation on Algorithmic Problem-Solving

Gabriel Souza Baggio, Gabriel Machado Lunardi, Guilherme Medeiros Machado, Subhankar Maity et al. · arXiv 2611.09791v1 · https://arxiv.org/abs/2611.09791v1 · relevance 0.38

TL;DR: Retrieval-augmented generation (RAG) using similarity-based example retrieval modestly improves LLM performance on competitive programming, raising pass@1 to 20.37% and Elo to 1014 versus weaker baselines.

Problem
Competitive programming demands not only code generation but also advanced algorithmic reasoning, posing distinctive challenges for large language models (LLMs) [1].

Method
The authors built a similarity-based RAG pipeline that embeds problem statements, searches a knowledge base of solved CodeContests problems, and retrieves relevant examples using a dynamic, query-dependent threshold to limit noise [1][2][4]. Three prompting conditions were compared: (i) zero-shot (no examples), (ii) random-retrieval (irrelevant examples), and (iii) similarity-based RAG (relevant examples) [1][2]. Experiments used the CodeContests dataset and focused on 81 test problems [4][6]. Six well-known LLMs were evaluated, and a Min-K membership-inference analysis was added to assess potential memorization of training data [2][5].

Key results
- On the 81-problem benchmark, the RAG condition achieved a pass@1 of 20.37%, outperforming zero-shot (17.9%) and random-retrieval (17.28%) [6][7].
- The 2.47-percentage-point gain over random retrieval demonstrates the benefit of contextually relevant examples, while random retrieval caused a 0.62% drop relative to zero-shot [6].
- In Elo terms, the RAG approach yielded an average rating of 1014, compared with 925 for zero-shot and 927 for random retrieval [7].
The study evaluated six LLMs across three prompting strategies on a subset of 81 CodeContests test problems with retrievable analogues [4][6].

Limitations
The evaluation was limited to a filtered subset of 81 problems that had at least one similar training example, so findings may not extend to the full CodeContests test set or to problems lacking close analogues [6][7]. Only six LLMs and a single dataset (CodeContests) were examined [2][4].

Sources

[1] arXiv 2611.09791v1.pdf, arXiv abstract

[2] arXiv 2611.09791v1.pdf, page 4

[4] arXiv 2611.09791v1.pdf, page 7

[5] arXiv 2611.09791v1.pdf, page 11

[6] arXiv 2611.09791v1.pdf, page 14

[7] arXiv 2611.09791v1.pdf, page 19

[P3] GRAC: Graph-Based Retrieval-Augmented Captioning for Surgical Scene Description

Rochak Dhakal, Chen Zhao, Soup Cobb, Xin Qian et al. · arXiv 2611.09866v1 · https://arxiv.org/abs/2611.09866v1 · relevance 0.38

TL;DR: GRAC is a graph-based, retrieval-augmented captioning system that outperforms prior surgical captioning models on standard benchmarks, but its gains rely on specific components and its evaluation is limited to the datasets and protocols used.

Problem
Existing captioning methods primarily learn surgical language patterns through model parameters despite the repetitive nature of surgical activities [2].

Method
GRAC encodes a scene graph of instruments, actions, and anatomical structures, retrieves up to three similar graph-caption pairs using a contrastive learning objective, and conditions caption generation on a comparison between the query and retrieved graphs via a graph-conditioned memory module [2][3][4].

Key results
- On EndoVis-18, GRAC attains a CIDEr-D of 5.88, compared with 4.23 for the SGT baseline and 3.36 for the strongest conventional Transformer baseline [1].
- Exact instrument-action-target triplet-set accuracy on EndoVis-18 reaches 52.6%, versus 32.7% for SGT and 26.2% for the strongest conventional baseline [1].
- On the TORS dataset, GRAC achieves a CIDEr-D of 5.06, compared with 3.80 for the strongest baseline [1].
- Ablation shows that retrieving three examples yields recall-at-1 of 0.621 and recall-at-3 of 0.855, while increasing to five examples does not improve quality [5].
- Removing both retrieval and interaction features drops CIDEr-D to 4.870; adding interaction edges alone raises it to 5.632; adding caption-only retrieval raises it to 5.805; full GRAC with compatibility supervision reaches 5.883, with exact triplet-set accuracy 0.526 and triplet F1 0.636 [5].
- Instrument hallucination decreases from 0.049 to 0.037 when compatibility supervision is added [5].
- Incorporating temporal context (adjacent frames) yields CIDEr-D 5.749 and exact triplet accuracy 0.514, slightly lower than the single-frame GRAC (5.883 and 0.526) [5].

Limitations
- Direct comparison with published SGT and SGT++ results is constrained by differing input features, evaluation procedures, and the lack of a public SGT++ implementation, limiting claims of superiority [7].
- Some improvements (e.g., compatibility supervision, graph comparison) are not statistically significant on primary metrics, indicating modest impact beyond baseline retrieval augmentation [5].
- Temporal context provides no clear benefit under the current experimental setting, and increasing the retrieval set beyond three examples does not further enhance performance [5].
- The evaluation is confined to the EndoVis-18 and TORS benchmarks; generalizability to other surgical domains remains untested.

Sources

[1] arXiv 2611.09866v1.pdf, arXiv abstract

[2] arXiv 2611.09866v1.pdf, page 3

[3] arXiv 2611.09866v1.pdf, page 6

[4] arXiv 2611.09866v1.pdf, page 9

[5] arXiv 2611.09866v1.pdf, page 11

[7] arXiv 2611.09866v1.pdf, page 17

[P4] SkinMind: Equipping Agentic RAG with Human-Inspired Graph Probing for Reliable Dermatology AI

Jing Hao, Weinan Zhou, Zhihong Lu, Yaling Shen et al. · arXiv 2611.08935v1 · https://arxiv.org/abs/2611.08935v1 · relevance 0.37

TL;DR: SkinMind is an agentic retrieval-augmented generation system that interactively probes a large multimodal dermatology knowledge graph to improve VQA performance, yielding ~8-point gains over direct answering.

Problem
Reliable dermatology AI requires verifiable evidence from heterogeneous textual and visual sources, but conventional RAG relies on one-shot retrieval from flat corpora, limiting adaptation of retrieval granularity, multi-hop reasoning, and multimodal integration [1].

Method
SkinMind introduces a multimodal agentic RAG framework where an orchestrator iteratively interacts with a graph environment using eight hierarchical graph-probing tools to identify entities, inspect node information, traverse relations, and retrieve text and image evidence [1][2]. The system constructs SkinMind-KG from a 62 M-token corpus of dermatology guidelines and textbooks plus 423 K images, yielding over 1.0 M nodes and 2.0 M edges that unify textual, relational, and visual evidence [1]. During inference the orchestrator maintains a state (query, optional image, interaction history), selects tools or a FINISH action, and compresses or summarizes context to stay within token limits (4,096 per observation, 16,384 overall) [2]. Retrieval configurations employ Qwen3-Embedding-4B for text and DermLIP for images, with policy-controlled per-tool budgets (default 5 text/graph items and 5 images) [6].

Key results
Across six dermatology VQA benchmarks, SkinMind improves the average score from 59.20 to 67.90 (+8.70 points) with GPT-5.4-mini and from 60.53 to 69.42 (+8.89 points) with Gemini-2.5-Flash, consistently outperforming direct answering and other RAG baselines [3]. Compared with LightRAG, it gains 5.75 points (GPT-5.4-mini) and 3.60 points (Gemini-2.5-Flash) [3]. Relative to strong standalone models, SkinMind (GPT-5.4-mini) scores 67.90, close to GPT-5.5's 68.08 and surpasses DermoGPT-RL by 1.36, SkinVL-PubMM by 16.35, and Lingshu-32B by 26.00 points [3]. Ablation on DDI and SNU134-4 shows full SkinMind achieves 65.09 and 85.00; removing graph exploration drops 2.44 and 2.50 points; removing image retrieval drops 1.83 and 5.83 points; removing progressive disclosure drops 0.61 and 4.17 points [3].

Limitations
The provided excerpts do not specify any limitations of the approach.

Sources

[1] arXiv 2611.08935v1.pdf, arXiv abstract

[2] arXiv 2611.08935v1.pdf, page 5

[3] arXiv 2611.08935v1.pdf, page 10

[6] arXiv 2611.08935v1.pdf, page 20

[P5] MARS: Training Multilingual Search Agents for Retriever-Free Evidence Acquisition

Rui Qi, Yunlong Liang, Yufeng Chen, Jinan Xu et al. · arXiv 2611.09090v1 · https://arxiv.org/abs/2611.09090v1 · relevance 0.33

TL;DR: MARS is a reinforcement-learning-based multilingual search agent that interacts directly with raw language-specific corpora via executable commands, achieving higher exact-match accuracy while

markedly cutting redundant searches compared with retrieval-augmented baselines.

Problem
Current multilingual retrieval-augmented generation depends on learned retrievers and similarity-based ranking, which are prone to cross-lingual retrieval errors and require costly indexing [1]. Moreover, exact search is fragile to entity surface-form mismatches and outcome-only supervision offers limited guidance for multi-turn search in multilingual settings [1].

Method
MARS frames multilingual search as direct corpus interaction (DCI) where the model generates reasoning traces and selects a target language before issuing executable shell commands [2]. It introduces corpus-native entity grounding that extracts high-confidence entity forms from passages and augments observations with explicit anchor tags, without external translators or linkers [3]. A redundancy-aware search reward penalizes near-duplicate commands (using Jaccard similarity) and rewards concise, successful trajectories, encouraging non-redundant evidence acquisition [3]. The whole system is trained with reinforcement learning to optimize these rewards [1][6].

Key results
MARS attains the highest exact-match (EM) scores among all evaluated baselines, achieving 40.64 % EM versus 37.07 % for the strongest DCI baseline (Grepseek) [5]. It reduces the average number of search commands from 4.22 to 2.56 (a 39 % drop) and the average number of interaction turns from 1.87 to 1.36 (27 % drop) [5]. Duplicate-search rate falls from 16.4 % to 4.9 % (~70 % reduction), and response token length shrinks from 4,853 to 3,447 tokens (~29 % reduction) [5]. Ablating the redundancy-aware reward raises duplicate rate to 22.33 % and increases commands to 5.51, confirming its impact [5]. Performance remains stable as corpus-language coverage expands, unlike a DCI baseline that degrades with more languages [5].

Limitations
MARS requires task-specific training data and incurs additional optimization cost, and its experiments cover only three tasks in a limited set of languages [7]. Direct corpus interaction stays sensitive to lexical variation when useful aliases are absent from the corpus, suggesting a need for hybrid retrieval approaches [7].

Sources

[1] arXiv 2611.09090v1.pdf, arXiv abstract

[2] arXiv 2611.09090v1.pdf, page 2

[3] arXiv 2611.09090v1.pdf, page 4

[5] arXiv 2611.09090v1.pdf, page 7

[6] arXiv 2611.09090v1.pdf, page 8

[7] arXiv 2611.09090v1.pdf, page 9